

# Cloud Computing Fundamentals

The shared vocabulary behind every AZ-104 topic — service models, regions, and how Azure organizes resources

 Beginner

 10 Minutes

 AZ-104 · Prerequisites

### Why This Matters

AZ-104 assumes you understand the basic vocabulary of cloud computing — service models, regions, and who is responsible for what. This shows up throughout the exam, especially in questions about availability, disaster recovery, and choosing between services.

## Service Models: IaaS, PaaS, SaaS

| Model                                     | What the provider manages                     | What you manage         | Azure examples                     |
|-------------------------------------------|-----------------------------------------------|-------------------------|------------------------------------|
| <b>IaaS</b> (Infrastructure as a Service) | Physical hardware, virtualization, networking | OS, runtime, apps, data | Virtual Machines, Virtual Networks |
| <b>PaaS</b> (Platform as a Service)       | Infrastructure + OS + runtime                 | Apps and data           | App Service, Azure SQL Database    |
| <b>SaaS</b> (Software as a Service)       | Everything, including the application         | Your data/config only   | Microsoft 365, Dynamics 365        |

**Example:** Running a web app on a VM you configure yourself (install IIS or nginx, patch the OS) is **IaaS**. Deploying the same app to **App Service** — where Azure handles the OS and runtime — is **PaaS**. AZ-104 covers both (Module 03 – Compute includes VMs *and* App Service).

## The Shared Responsibility Model

As you move from IaaS → PaaS → SaaS, Microsoft takes on more responsibility and you take on less. But **security of your data, identities, and access management is always your responsibility**, regardless of model. This is why Identity & Governance (Module 01) is weighted so heavily on the exam — it's the part you always own.

## Regions, Region Pairs & Availability Zones

- A **region** is a set of Microsoft datacenters in a geographic area (e.g., *East US*). Most resources are deployed to a specific region.
- An **availability zone (AZ)** is a physically separate datacenter (with independent power, cooling, networking) *within* a region. Regions that support AZs typically have 3.
- A **region pair** is a pre-defined pairing of two regions in the same geography (e.g., *East US* ↔ *West US*) used for disaster recovery — services like geo-redundant storage (GRS) replicate to the paired region.

**Example:** Deploying VMs across **3 availability zones** in *East US* protects against a datacenter-level failure (power/cooling outage in one building). Enabling **GRS** on a storage account in *East US* additionally protects against a *regional* disaster by replicating data to *West US*.

You'll see AZs again in Module 03 (VM availability) and region pairs again in Module 02 (storage redundancy) and Module 05 (Azure Site Recovery).

### East US (Region)

- ├─ Availability Zone 1 (Datacenter A)
  - ├─ Availability Zone 2 (Datacenter B)
  - └─ Availability Zone 3 (Datacenter C)
- Protects against datacenter failure

### Region Pair

East US ↔ West US

GRS replication target

Protects against regional disaster

# Resource Hierarchy (Preview)

Azure organizes everything under a hierarchy:

```
Management Group
├── Subscription
│   ├── Resource Group
│       └── Resource (VM, storage account, VNet, ...)
```

Module 01 covers this hierarchy in depth (management groups, subscriptions, resource groups, and how policies/RBAC apply at each level). For now, just know that every resource you create in these labs lives inside a **resource group** — which is why every lab's cleanup step is "delete the resource group."

## How This Connects to Other Topics

### Glossary

Unfamiliar term? Check the [AZ-104 Glossary](#) for quick definitions of every concept used across these docs.

### Module 01 — Identity & Governance

The resource hierarchy above (management groups → subscriptions → resource groups) is covered in depth in [Management Groups & Azure Policy](#).

## Key Takeaways

#### 💡 Summary

- **IaaS / PaaS / SaaS** = how much Microsoft manages vs. you
- **Shared responsibility** = security of data & identity is always yours
- **Region** = a geographic area of datacenters

- **Availability Zone** = a physically separate datacenter within a region
- **Region pair** = two regions used together for disaster recovery (e.g., GRS)
- **Resource hierarchy** = Management Group → Subscription → Resource Group → Resource

# Next Steps

1. **Learn:** Read this doc (you're here)
2. **Continue:** Read **Networking Basics** (IP addressing, subnets, DNS — needed for Module 04)
3. **Continue:** Read **Identity & Access Fundamentals** (AuthN vs. AuthZ — needed for Module 01)
4. **Begin: Module 01: Entra ID Overview** — the first hands-on module

| Azure AZ-104                       | Module             | Generated     |
|------------------------------------|--------------------|---------------|
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